



WEEK 3 DOCUMENT

Automated Incident Root Cause Analysis

(Model Training using SVM & Logistic Regression)

◆ 1. Introduction

In Week 3, the focus of the project was on **training and evaluating machine learning models** to predict the **root cause (close code)** of IT incidents.

Instead of using multiple complex models, we focused on:

- Logistic Regression
- Support Vector Machine (SVM)

This approach ensures **simplicity, interpretability, and efficiency**.

◆ 2. Objective

The objectives of Week 3 are:

- Train two classification models
 - Compare their performance
 - Select the best model
 - Save the model for future use
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◆ 3. Dataset Used

We used the cleaned dataset:

👉 `cleaned_incidents.csv`

✓ Dataset Features:

- Preprocessed and cleaned (Week 1)
 - No missing values
 - Encoded categorical data
 - Target variable: **close_code**
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◆ 4. Problem Type

This is a **Multi-Class Classification Problem**:

- Input → Incident features
 - Output → Close Code
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◆ 5. Models Used

◆ 5.1 Logistic Regression

- Linear classification algorithm
 - Fast and efficient
 - Easy to interpret
 - Works well for structured data
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◆ 5.2 Support Vector Machine (SVM)

- Finds optimal decision boundary
 - Works well in high-dimensional spaces
 - Effective for classification tasks
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◆ 6. Training Process

- Dataset split:

- 80% Training
- 20% Testing

- ✓ Both models trained on same dataset
 - ✓ Ensures fair comparison
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◆ 7. Evaluation Metrics

Models were evaluated using:

- Accuracy
- Precision
- Recall
- **F1-score (important)**

👉 F1-score is used for **balanced evaluation**

◆ 8. Model Comparison

Example comparison:

Model	Accuracy	F1 Score
Logistic Regression	0.75	0.74
SVM	0.78	0.77

◆ 9. Best Model Selection

👉 Model selected based on:

- Higher **F1-score**
- Better generalization

✓ **Selected Model:**

SVM / Logistic Regression (based on your results)

◆ 10. Visualization



Model Comparison Graph

- Bar chart comparing Accuracy & F1-score



Confusion Matrix

- Shows prediction performance
 - Identifies errors
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◆ 11. Model Saving

The best model was saved using:

```
joblib.dump(best_model, 'best_model.pkl')
```

◆ 12. Outcome of Week 3

- ✓ Trained two ML models
 - ✓ Compared performance
 - ✓ Selected best model
 - ✓ Saved model for deployment
 - ✓ Generated evaluation visuals
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◆ 13. Conclusion

Week 3 successfully built a **reliable classification system** using simple yet effective models.

This step ensures:

- Good prediction performance

- Easy explainability
 - Strong foundation for deployment
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VIVA QUESTIONS (UPDATED)

◆ Basic

1. What is Logistic Regression?

👉 A linear model used for classification that predicts probability.

2. What is SVM?

👉 A model that finds the best boundary (hyperplane) to separate classes.

3. What type of problem is this?

👉 Multi-class classification.

◆ Intermediate

4. Why use Logistic Regression?

👉 It is simple, fast, and interpretable.

5. Why use SVM?

👉 It performs well in high-dimensional data and complex boundaries.

6. Why use F1-score instead of accuracy?

👉 Because it balances precision and recall.

7. What is overfitting?

👉 When a model performs well on training data but poorly on new data.

◆ Advanced

8. Difference between Logistic Regression and SVM?

👉 Logistic Regression predicts probabilities, while SVM focuses on margin maximization.

9. What is a hyperplane?

👉 A boundary that separates different classes.

10. Why save the model?

👉 To reuse it without retraining.